

HFB densities

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Abstract

Basic relations used for computation of various densities.

1 General case

HFB problem in general case is given by:

$$\begin{pmatrix} h_{\uparrow\uparrow} & h_{\uparrow\downarrow} & 0 & \Delta \\ h_{\downarrow\uparrow} & h_{\downarrow\downarrow} & -\Delta & 0 \\ 0 & -\Delta^* & -h_{\uparrow\uparrow}^* & -h_{\uparrow\downarrow}^* \\ \Delta^* & 0 & -h_{\downarrow\uparrow}^* & -h_{\downarrow\downarrow}^* \end{pmatrix} \begin{pmatrix} u_{n\uparrow} \\ u_{n\downarrow} \\ v_{n\uparrow} \\ v_{n\downarrow} \end{pmatrix} = E_n \begin{pmatrix} u_{n\uparrow} \\ u_{n\downarrow} \\ v_{n\uparrow} \\ v_{n\downarrow} \end{pmatrix} \quad (1)$$

and corresponding densities are constructed as follow:

$$n_\sigma(\mathbf{r}) = \sum_{|E_n| < E_c} |v_{n,\sigma}(\mathbf{r})|^2 f_\beta(-E_n), \quad (2)$$

$$\tau_\sigma(\mathbf{r}) = \sum_{|E_n| < E_c} |\nabla v_{n,\sigma}(\mathbf{r})|^2 f_\beta(-E_n), \quad (3)$$

$$\nu(\mathbf{r}) = \frac{1}{2} \sum_{|E_n| < E_c} (u_{n,\uparrow}(\mathbf{r})v_{n,\downarrow}^*(\mathbf{r}) - u_{n,\downarrow}(\mathbf{r})v_{n,\uparrow}^*(\mathbf{r})) f_\beta(-E_n), \quad (4)$$

$$\mathbf{j}_\sigma(\mathbf{r}) = \sum_{|E_n| < E_c} \text{Im}[v_{n,\sigma}(\mathbf{r}) \nabla v_{n,\sigma}^*(\mathbf{r})] f_\beta(-E_n), \quad (5)$$

where subscripts $\sigma = \{\uparrow, \downarrow\}$ indicate spin, E_n denotes quasi-particle energy and E_c is energy cut-off scale. Fermi distribution function $f_\beta(E) = 1/(\exp(\beta E) + 1)$ is introduced to model temperature $k_B T = 1/\beta$ effects. In zero temperature limit $T = 0$ Fermi distribution function reduces to $f_\beta(E) = \theta(-E)$, where $\theta(x)$ is Heaviside step function. Note also that $f_\beta(E) = 1 - f_\beta(-E)$.

Relation 1: If vector $\varphi_+ = (u_{n\uparrow}, u_{n\downarrow}, v_{n\uparrow}, v_{n\downarrow})^T$ is solution of Eq. (1) with eigenvalue E_n , then also vector $\varphi_- = (v_{n\uparrow}^*, v_{n\downarrow}^*, u_{n\uparrow}^*, u_{n\downarrow}^*)^T$ is solution with eigenvalue $-E_n$.

Conclusion 1: It is sufficient to extract only solutions for positive eigenvalues E_n .

2 Case where $h_{\uparrow\downarrow} = h_{\downarrow\uparrow} = 0$

If $h_{\uparrow\downarrow} = h_{\downarrow\uparrow} = 0$ (typically it corresponds to no spin-orbit case) then Eq. (1) decouples into two sets of equations:

$$\begin{pmatrix} h_{\uparrow\uparrow} & \Delta \\ \Delta^* & -h_{\downarrow\downarrow}^* \end{pmatrix} \begin{pmatrix} u_{n\uparrow} \\ v_{n\downarrow} \end{pmatrix} = E_n \begin{pmatrix} u_{n\uparrow} \\ v_{n\downarrow} \end{pmatrix}, \quad (6)$$

$$\begin{pmatrix} h_{\downarrow\downarrow} & -\Delta \\ -\Delta^* & -h_{\uparrow\uparrow}^* \end{pmatrix} \begin{pmatrix} u_{n\downarrow} \\ v_{n\uparrow} \end{pmatrix} = E_n \begin{pmatrix} u_{n\downarrow} \\ v_{n\uparrow} \end{pmatrix}. \quad (7)$$

Relation 2: If vector $\varphi_+ = (u_{n\uparrow}, v_{n\downarrow})^T$ is solution of Eq. (6) with eigenvalue E_n , then vector $\varphi_- = (v_{n\uparrow}^*, u_{n\downarrow}^*)^T$ is solution of Eq. (7) with eigenvalue $-E_n$.

Conclusion 2: Having solutions of Eq. (6) one can construct solutions of Eq. (7) by using transformation: $u_{n,\uparrow} \rightarrow v_{n,\uparrow}^*$, $v_{n,\downarrow} \rightarrow u_{n,\downarrow}^*$ and $E_n \rightarrow -E_n$. In practice we solve only Eq. (6) and we express all densities via $\{u_{n\uparrow}, v_{n\downarrow}\}$:

$$n_{\uparrow}(\mathbf{r}) = \sum_{|E_n| < E_c} |u_{n,\uparrow}(\mathbf{r})|^2 f_{\beta}(E_n), \quad (8)$$

$$n_{\downarrow}(\mathbf{r}) = \sum_{|E_n| < E_c} |v_{n,\downarrow}(\mathbf{r})|^2 f_{\beta}(-E_n) \quad (9)$$

$$\tau_{\uparrow}(\mathbf{r}) = \sum_{|E_n| < E_c} |\nabla u_{n,\uparrow}(\mathbf{r})|^2 f_{\beta}(E_n), \quad (10)$$

$$\tau_{\downarrow}(\mathbf{r}) = \sum_{|E_n| < E_c} |\nabla v_{n,\downarrow}(\mathbf{r})|^2 f_{\beta}(-E_n), \quad (11)$$

$$\nu(\mathbf{r}) = \frac{1}{2} \sum_{|E_n| < E_c} u_{n,\uparrow}(\mathbf{r}) v_{n,\downarrow}^*(\mathbf{r}) (f_{\beta}(-E_n) - f_{\beta}(E_n)), \quad (12)$$

$$\mathbf{j}_{\uparrow}(\mathbf{r}) = - \sum_{|E_n| < E_c} \text{Im}[u_{n,\uparrow}(\mathbf{r}) \nabla u_{n,\uparrow}^*(\mathbf{r})] f_{\beta}(E_n), \quad (13)$$

$$\mathbf{j}_{\downarrow}(\mathbf{r}) = \sum_{|E_n| < E_c} \text{Im}[v_{n,\downarrow}(\mathbf{r}) \nabla v_{n,\downarrow}^*(\mathbf{r})] f_{\beta}(-E_n), \quad (14)$$

$$S = -k_B \sum_{|E_n| < E_c} (f_{\beta}(E_n) \ln f_{\beta}(E_n) + f_{\beta}(-E_n) \ln f_{\beta}(-E_n)). \quad (15)$$

3 Case where $h_{\uparrow\uparrow} = h_{\downarrow\downarrow} = h$

3.1 For any temperature $T \geq 0$

If $h_{\uparrow\uparrow} = h_{\downarrow\downarrow} = h$ (typically it corresponds to the spin-balanced case) then Eq. (6) takes form, as typically cited in papers (spin indices are dropped):

$$\begin{pmatrix} h & \Delta \\ \Delta^* & -h^* \end{pmatrix} \begin{pmatrix} u_n \\ v_n \end{pmatrix} = E_n \begin{pmatrix} u_n \\ v_n \end{pmatrix}. \quad (16)$$

Relation 3: If vector $\varphi_+ = (u, v)^T$ is solution of Eq. (16) with eigenvalue E_n , then vector $\varphi_- = (v_n^*, -u_n^*)^T$ is also solution with eigenvalue $-E_n$.

Conclusion 3: Having solutions of Eq. (16) for positive energies E_n one can construct all solutions of this equation.

In practice for spin-balanced case we solve Eq. (16) for positive E_n and we express all densities via these states:

$$n_{\uparrow}(\mathbf{r}) = n_{\downarrow}(\mathbf{r}) = \sum_{0 < E_n < E_c} (|v_n(\mathbf{r})|^2 f_{\beta}(-E_n) + |u_n(\mathbf{r})|^2 f_{\beta}(E_n)) \quad (17)$$

$$\tau_{\uparrow}(\mathbf{r}) = \tau_{\downarrow}(\mathbf{r}) = \sum_{0 < E_n < E_c} (|\nabla v_n(\mathbf{r})|^2 f_{\beta}(-E_n) + |\nabla u_n(\mathbf{r})|^2 f_{\beta}(E_n)), \quad (18)$$

$$\nu(\mathbf{r}) = \sum_{0 < E_n < E_c} u_n(\mathbf{r}) v_n^*(\mathbf{r}) (f_{\beta}(-E_n) - f_{\beta}(E_n)), \quad (19)$$

$$\begin{aligned} \mathbf{j}_{\uparrow}(\mathbf{r}) = \mathbf{j}_{\downarrow}(\mathbf{r}) = & \sum_{0 < E_n < E_c} (\text{Im}[v_n(\mathbf{r}) \nabla v_n^*(\mathbf{r})] f_{\beta}(-E_n) \\ & - \text{Im}[u_n(\mathbf{r}) \nabla u_n^*(\mathbf{r})] f_{\beta}(E_n)), \end{aligned} \quad (20)$$

$$\begin{aligned} S = & -2k_B \sum_{0 < E_n < E_c} (f_{\beta}(E_n) \ln f_{\beta}(E_n) \\ & + f_{\beta}(-E_n) \ln f_{\beta}(-E_n)). \end{aligned} \quad (21)$$

3.2 For $T = 0$

For $T = 0$ (and therefore $\beta \rightarrow \infty$) the value of $f_{\beta}(\pm E)$ sets to fixed values:

$$f_{\beta}(+E) \xrightarrow{\beta \rightarrow \infty} \frac{1}{e^{+E \cdot \infty} + 1} = 0, \quad (22)$$

$$f_{\beta}(-E) \xrightarrow{\beta \rightarrow \infty} \frac{1}{e^{-E \cdot \infty} + 1} = 1. \quad (23)$$

$f_{\beta}(+E)$ converges to 0 from the positive numbers' side, resulting in

$$f_{\beta}(+E) \ln f_{\beta}(+E) \xrightarrow{f_{\beta}(+E) \rightarrow 0^+} 0. \quad (24)$$

The equations of densities change to:

$$n_{\uparrow}(\mathbf{r}) = n_{\downarrow}(\mathbf{r}) = \sum_{0 < E_n < E_c} |v_n(\mathbf{r})|^2 \quad (25)$$

$$\tau_{\uparrow}(\mathbf{r}) = \tau_{\downarrow}(\mathbf{r}) = \sum_{0 < E_n < E_c} |\nabla v_n(\mathbf{r})|^2, \quad (26)$$

$$\nu(\mathbf{r}) = \sum_{0 < E_n < E_c} u_n(\mathbf{r}) v_n^*(\mathbf{r}), \quad (27)$$

$$\mathbf{j}_{\uparrow}(\mathbf{r}) = \mathbf{j}_{\downarrow}(\mathbf{r}) = \sum_{0 < E_n < E_c} \text{Im}[v_n(\mathbf{r}) \nabla v_n^*(\mathbf{r})], \quad (28)$$

$$S = 0. \quad (29)$$